FARDIN IMRAN SHAIKH

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PROFESSIONAL SUMMARY

Software and IoT engineer with 3+ years of experience independently handling both the hardware and software sides of connected-device solutions from embedded hardware, wiring, and sensor integration (ESP32, Raspberry Pi) through backend APIs, cloud services, and Flutter mobile/web applications. Skilled in designing full-stack apps that let users monitor and control IoT hardware in real time, including live sensor dashboards, remote device control, and cloud-based data pipelines using AWS IoT Core, Lambda, DynamoDB, Firebase, Supabase, and Render. Also experienced applying AI/ML models (YOLOv8, TFLite, MobileNetSSD) for edge-device detection use cases, and using AI-assisted development tools to speed up building and debugging. Have independently guided and delivered 50+ end-to-end IoT application builds (mobile app + backend + database + hardware). Strong foundation in Python, SQL, REST APIs, and system integration comfortable working across the complete stack of a connected product, and open to Software Engineer as well as IoT Application Engineer roles.

TECHNICAL SKILLS

- **IoT & Embedded Systems:** ESP32, Raspberry Pi, Arduino, Sensors & Actuators, Sensor Integration, Hardware-Software Integration, IoT Prototyping, Real-Time Device Monitoring & Control
- **IoT Protocols & Connectivity:** MQTT, HTTP/HTTPS, TCP/IP, Wi-Fi, I2C, SPI, UART, LoRa
- **IoT Platforms & Dashboards:** Node-RED, Mosquitto, ThingsBoard, ThingSpeak, Grafana, Real-Time IoT Dashboards
- **Cloud & IoT Services:** AWS IoT Core, AWS Lambda, Amazon DynamoDB, Firebase (Authentication, Firestore, Realtime Database), Supabase, Render, Azure IoT Hub
- **App & Software Development:** Flutter (Mobile & Web), Dart, Python, REST API Design & Integration, JSON, Git/GitHub
- **Backend & Databases:** MySQL, SQL, REST APIs, Database Design, Cloud Data Storage, Data Cleaning & Validation
- **Data Analytics & Visualization:** Power BI, Pandas, NumPy, Excel, Data Analysis, Data Visualization, Real-Time KPI Dashboards
- **AI, Computer Vision & Edge AI:** YOLOv8, TensorFlow Lite (TFLite), MobileNetSSD, Object Detection, Edge AI, AI-Assisted Development Tools
- **Development & Engineering Tools:** VS Code, Arduino IDE, Jupyter Notebook, Postman, Linux/Raspberry Pi OS, Figma, 3D Modelling & Prototyping

PROFESSIONAL EXPERIENCE

IoT Development Engineer and Application Trainer

April 2024 – Present

Grok Learning Pvt Ltd | Mumbai, Maharashtra, India

- Build full-stack IoT applications using Flutter (Android & Web) for the frontend, Python/REST APIs for the backend, and MySQL for data storage, integrated with ESP32 and Raspberry Pi hardware over MQTT/HTTP for real-time device monitoring and control.
- Develop cloud-connected device pipelines using AWS IoT Core, AWS Lambda, and Amazon DynamoDB, alongside Firebase, Supabase, and Render, to stream live sensor data from embedded hardware to mobile and web dashboards.
- Design and deploy real-time monitoring dashboards using Node-RED, ThingsBoard, and Power BI, translating raw sensor data (temperature, humidity, air quality, soil/water metrics) into live KPIs and automated alerts.
- Built an IoT Weather Station on Raspberry Pi with MQ7, DHT11, PM2.5, and rain sensors, paired with a Flutter Android/web app and a Python backend on Render/Supabase for fully cloud-based, real-time weather monitoring.
- Built an IoT-based Biogas Monitoring System on Raspberry Pi handling hardware/component selection, sensor data acquisition, REST API integration, and a live web dashboard for real-time visualization.
- Apply Python and REST API integration to connect embedded sensors, cloud databases, and mobile/web applications into a single working product, debugging and resolving hardware-software integration issues across the stack.
- Document technical architecture and project specifications, and coordinate with cross-functional stakeholders to gather requirements and deliver on schedule.

Robotics Project Engineer *June 2023 – March 2024*

Atal Innovation Mission (NITI Aayog – Hope Foundation) | *Mumbai*

- Managed end-to-end execution of 15+ robotics and embedded-systems projects, coordinating timelines, resources, and stakeholder expectations with 100% on-time completion.
- Built robotics prototypes incorporating sensor data collection, embedded programming, and hardware-software integration.
- Developed technical documentation and structured workflows for concurrent project builds, tracking milestones and quality outcomes.

KEY PROJECTS

IoT Weather Station | *Raspberry Pi, Flutter, Python, Render, Supabase*

- Built a cloud-based IoT weather station using Raspberry Pi as the central controller, wiring and integrating four sensors MQ7 (air pollution), DHT11 (temperature & humidity), PM2.5 (particulate matter), and a rain sensor for real-time environmental monitoring, housed in a custom 3D-designed and printed enclosure.
- Developed a Python backend on Raspberry Pi to read sensor data and sync it to Supabase, with Render handling API hosting enabling fully cloud-based data flow instead of relying on a local network connection.
- Built the companion Android and web dashboard app, deployed on Firebase Hosting, pulling live sensor data from the cloud so users can check real-time weather and air-quality conditions from anywhere, on any device.

AgriConnect – Smart Farming IoT Platform | *ESP32, Raspberry Pi, MQTT (Mosquitto), Node-RED, SQLite, ThingSpeak*

- Built an automated irrigation and monitoring system using ESP32 to read soil moisture, temperature, and humidity (DHT11) and publish live data to a Raspberry Pi running a Mosquitto MQTT broker.
- Implemented automatic pump control based on soil-moisture thresholds, plus manual pump ON/OFF control via a Node-RED dashboard, using bidirectional MQTT communication between the ESP32 and Raspberry Pi. Logged sensor data locally in SQLite on the Raspberry Pi and streamed it to ThingSpeak for cloud-based historical analytics, with a live Node-RED dashboard for real-time gauges and time-series graphs.

NestHawk – AI-Powered Edge Detection System | *Python, YOLOv8-TFLite, MobileNetSSD, Raspberry Pi, SQL*

- Built a real-time bird detection system for a residential building's private pool, where pigeons were repeatedly fouling the pool area and disturbing residents deployed YOLOv8-TFLite and MobileNetSSD on Raspberry Pi edge hardware with a camera to detect birds in real time, achieving 95%+ accuracy with sub-200ms response time.
- Designed an automated deterrent trigger that activates sprinklers and ultrasonic sound waves the moment a bird is detected near the pool, driving pigeons away without harming them or requiring manual intervention.

ESP32 to AWS IoT Cloud Pipeline | *ESP32, AWS IoT Core, AWS Lambda, DynamoDB, MQTT, DHT11*

- Provisioned an AWS IoT Core “Thing” with device certificates/policy and programmed an ESP32 to publish live DHT11 sensor data over MQTT; configured an IoT Core Rule to trigger AWS Lambda, writing readings into DynamoDB a complete device-to-cloud pipeline (ESP32 → IoT Core → Lambda → DynamoDB).

IoT-Based Biogas Plant Monitoring & Control System | *Raspberry Pi, Python, ESP32, API, Web Dashboard*

- Built end-to-end IoT monitoring system for biogas digester selected sensors/actuators, wired circuits, and developed firmware/software independently.
- Integrated temperature, pH, methane, level, and weight sensors with Raspberry Pi/ESP32 for real-time monitoring and automated stirrer/valve/fan control. Developed a web dashboard for live remote monitoring of sensor data and system status.

EcoClaw – AI-Powered Autonomous Waste Collection Robot | *Raspberry Pi, MobileNetSSD V2, Flutter, 3D Printing/Design*

- Built an autonomous and manually-controlled waste collection robotic car with an integrated robotic arm, using Raspberry Pi as the controller and a custom-trained MobileNetSSD V2 model (trained from scratch on a self-collected image dataset) for real-time waste detection.
- Designed a Flutter app to switch between manual and autonomous modes, calibrated and integrated the robotic arm for pickup, and 3D-designed/printed the car chassis and arm components from scratch.

EDUCATION

Bachelor of Science in Information Technology | CGPA: 7.79 / 10
Smt. K.G. Mittal Institute of Management, IT & Research, Mumbai | 2020 – 2023

Higher Secondary Certificate (HSC) - Science & Electronics
Shri T.P. Bhatia Jr. College of Science, Mumbai | Completed March 2019